REQUIREMENTS MODELING

COURSE UNIT: SOFTWARE PRINCIPLES 2

COURSE : BSE

YEAR : ONE

SEMESTER : TWO

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FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

FUNCTIONAL REQUIREMENTS

Functional Requirements describe what the system must do. They specify particular behaviours, functions, or features the system should perform under specific conditions. In essence, they answer: "What tasks must the system carry out?" Examples include recording a transaction, generating a receipt, or calculating a daily summary.

Functional Requirements (FR-01 to FR-07)

FR-01 – Transaction Recording

The system shall allow the user to record a new transaction by entering customer name, item name, transaction type (sale or rental), quantity, unit price, and total amount paid. All mandatory fields must be completed. A unique sequential transaction ID and timestamp are assigned. If any mandatory field is empty or amount paid is zero/negative, an error is shown and the transaction is not saved.

FR-02 – Automated Receipt Generation

The system shall automatically generate a numbered receipt for every successfully saved transaction. The receipt must display the shop name, receipt number, date, customer name, item, transaction type, amount paid, and any balance due. The receipt appears on screen within 3 seconds, and a print option is available.

FR-03 – Expenditure Recording

The system shall allow the user to record business expenditures separately from sales income. Each expenditure record includes a category (selectable from a dropdown: Tailor Fees, Food, Delivery, Other), amount, and date. Expenditures are saved to a separate expenditure log with a unique ID and timestamp. Invalid categories or non-positive amounts trigger an error.

FR-04 – Daily Financial Summary

The system shall calculate and display a daily financial summary for any selected date. The summary shows total income from sales and rentals, total expenditure, and net balance (income minus expenditure). If no records exist for that date, a message is shown and net balance is displayed as UGX 0.

FR-05 – Inventory Management

The system shall maintain an inventory list of stock items, each with a name, quantity, and status (Available, Rented Out, or Sold). When a sale or rental transaction is recorded, the system automatically updates the item's quantity and status. For rentals, a manual return action resets status to Available. The system prevents transactions for items with zero available quantity.

FR-06 – Transaction Search and Dispute Resolution

The system shall allow the user to search the transaction log by customer name (partial match allowed), date (calendar picker), or receipt number (exact match). Matching records are displayed with full details to support resolution of customer payment disputes.

FR-07 – Item Price Calculator

The system shall provide a price calculation tool that computes a recommended selling price for a new item based on its import cost components (purchase cost, airfreight cost, importation tax) and a user-defined profit margin percentage. The formula is: total cost × (1 + margin/100), rounded to the nearest 1,000 UGX. The recommended price can be accepted or overridden.

NON FUNCTIONAL REQUIREMENTS

Non-Functional Requirements describe how well the system performs its functions. They address quality attributes such as usability, performance, reliability, security, and portability. They answer: “How should the system behave?” rather than what it does.

Non-Functional Requirements (NFR-01 to NFR-07)

NFR-01 – Usability

The system shall present all menus, labels, and prompts in plain English without technical jargon. A non-technical user must be able to complete a full transaction cycle (record → receipt) without assistance after a single 30-minute demonstration. Acceptance: the user completes a transaction and generates a receipt within 3 minutes of first independent use, in 3 out of 3 test attempts.

NFR-02 – Performance

The system shall generate and display a receipt within 3 seconds of the user confirming a transaction, measured on standard low-cost hardware (Intel Core i3 or equivalent, 4GB RAM, Windows 10). Acceptance: receipt is displayed within 3 seconds in 100% of tested transactions on the specified hardware.

NFR-03 – Reliability

The system shall not lose any recorded transaction or expenditure data if the application is closed unexpectedly, the computer is shut down without warning, or a power interruption occurs mid-session. Acceptance: all transactions recorded in a session are fully retrievable after application restart in 100% of tested cases, including simulated abrupt shutdown.

NFR-04 – Data Integrity

The system shall prevent the saving of any transaction or expenditure record with a zero, negative, or non-numeric amount by displaying a specific error message and blocking the save action. Acceptance: zero, negative, and non-numeric entries are rejected with a descriptive error in 100% of tested cases; no invalid record is persisted.

NFR-05 – Security

The system shall require the user to enter a 4-digit PIN before accessing any transaction records, financial summaries, or inventory data. Unauthenticated

access to any data screen is blocked. The system locks after 3 consecutive incorrect PIN entries and requires a restart to retry.

NFR-06 – Maintainability

The system code shall be commented such that at least 80% of all methods contain a descriptive comment explaining the method's purpose, its input parameters, and its return value. Acceptance: static code review confirms that 80% or more of all methods contain compliant comments before Phase 4 submission.

NFR-07 – Portability

The system shall be installable and fully operational on any Windows 10 or Windows 11 PC without requiring additional software installations beyond the .NET runtime. Acceptance: the system installs and runs successfully on clean Windows 10 and Windows 11 machines with no missing dependency errors.

CONTEXT DATA FLOW DIAGRAM OF A BOUTIQUE MANAGEMENT SYSTEM

(CONTEXT DFD)

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It illustrates the sources and destinations of data (external entities), the processes that transform data, and the data stores where data is held. DFDs are used to visualise system boundaries, inputs, outputs, and the flow of information between components without showing programming logic or control flows.

Context Diagram (Level 0 DFD)

The context diagram is the highest level (Level 0) of a DFD. It represents the entire system as a single process, showing only the external entities that interact with the system and the data flows that cross the External Entities

1. Customer

The Customer is the person who:

Provides transaction details (name, item, payment)

Receives a receipt after purchase or rental

2. Admin (Owner/Employee)

The Admin is responsible for:

- Managing products and inventory
- Recording expenditures
- Requesting reports and summaries

Main Process

Boutique Management System

This is the central system that:

- Processes all business operations
- Stores and manages data
- Generates outputs such as receipts and reports

Data Flows

1. Transaction Details (Customer → System)

- Includes customer name, item purchased/rented, quantity, and payment
- This data is used to create a transaction record

2. Receipt / Order Status (System → Customer)

After processing the transaction, the system generates:

- A receipt
- Payment confirmation

3. Product Info / Updates (Admin → System)

Admin sends:

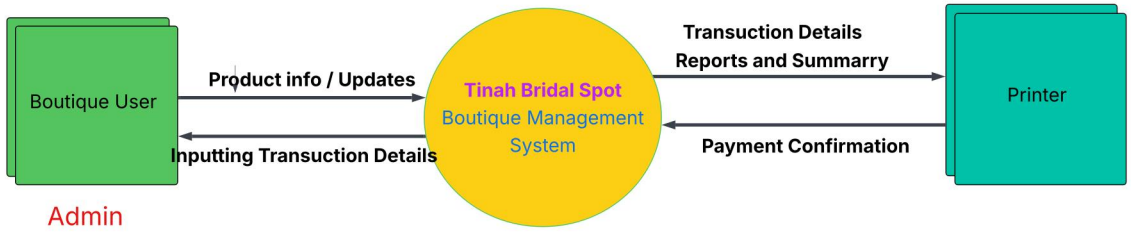
- New product details
- Price updates
- Inventory changes

4. Reports / Financial Summary (System → Admin)

The system provides:

Daily income

Expenditures



LEVEL 1 DATA FLOW DIAGRAM

The Level 1 DFD breaks down the main system into smaller functional processes based on the SRS document. It includes most of the entities that take part inside the system or internally

Main Processes

1. Record Transaction

Function:

- Captures customer purchase or rental details

Input:

- Customer name
- Item name
- Quantity
- Amount paid

Output:

- Transaction saved into Transaction Records

2. Generate Receipt

Function:

- Automatically creates a receipt after a transaction

Input:

Data from Transaction Records

Output:

- Receipt given to customer
- Includes receipt number, date, and payment details

3. Manage Inventory

Function:

- Handles stock management

Input:

- Product updates from Admin
- Sales data from transactions

Output:

- Updates Inventory Database

4. Record Expenditure

Function:

- Records business expenses

Input:

- Expense category (e.g., transport, food)
- Amount spent

Output:

- Stored in Expenditure Records

5. Generate Reports

Function:

- Produces financial summaries

Input:

- Transaction Records
- Expenditure Records

Output:

- Daily financial report:
- Total income
- Total expenditure
- Net balance

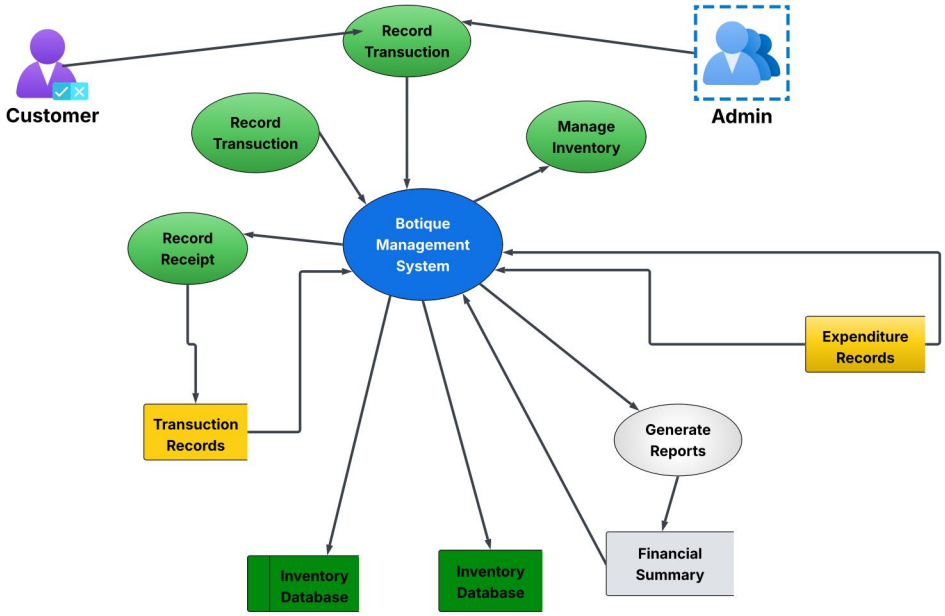
Data Stores

1. Transaction Records

Stores all sales and rental transactions

2. Inventory Database

Stores product details and stock levels

3. Expenditure Records

Stores all business expenses